(pumping lemma for regular Languages)

↘ proving certain languages are not regular.

If a lang satisfy Pumping lemma then the lang is Regular

Theorem

→ Let L be a regular Language (DFA)

→ Then there exist a constant n (no of states of dfa) such that for every string w in L.

$|w| \geq n$

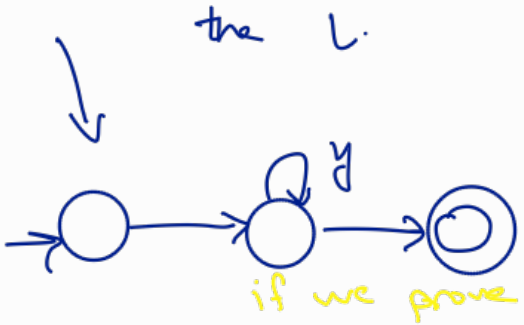
Chose a string from L → w and → $|w| \geq n$

Breaking $w = xyz$ → into 3 strings such that

1) $y \neq \varepsilon$

2) $|xy| \leq n$

3) For all $k \geq 0$, the string $xy^k z$ is also in the L.



if we prove all 3 then the Language is Regular

e.g

$L = \{ 0^n 1^n \mid n \geq 1 \}$

Assume L is Regular

Let n be no of states of dfa.

Let $w = 0^n 1^n$ $|w| \geq n$

Splitting xyz such that

$w = 0^n 1^n$ $n \geq 1$

$0^2 1^2$

0011

1) $y \neq \varepsilon$

2) $|xy| \leq n$

3) For all $k \geq 0$, $xy^k z \in L$

e.g $a^n b^{2n}$ $n \geq 0$

$\underbrace{aa}_{x}\overbrace{bbbb}^{y}\underbrace{}_{z}$ Dividing in xy^2 (your choice)

Pumping y RN

→ aa (bb) bb

→ aa $(bb)^k$ bb let k=2

→ aa (bbbb) bb → Doesn't belong.

k=0

aa $(bb)^0$ bb

aa (λ) bb

aabb

k=1

aa $(bb)^1$ bb

aabbbb

k=2

aa $(bb)^2$ bb

aabbbbbb

$L = 0^n 1^n \mid n \geq 1$

(too short) 01 n=1

0011 n=2

000111 n=3

00001111 n=4

picking this for xy^2.

Testing $\underbrace{00}_{x}\overbrace{1}^{y}\underbrace{1}_{z}$ $y \neq \varepsilon$ ✓

i=2 Pumping

00111

$\notin L$

hence proved not regular

Note xyz can be any substring but its order must not change.

e.g Prove that $L = \{ a^i b^j \mid i \leq j \}$ is not regular

$i = 0 \quad j = 0$

$a^0 b^0 \rightarrow \wedge$

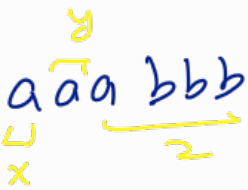
$i = 0 \quad j = 1$

$a^0 b^1 \rightarrow b$

min I need three strings for xyz

$i = 3 \quad j = 3$

$a^3 b^3 \rightarrow aaabbb$



$y \neq \epsilon$ ✓

$K = 2$

$a(a)^2(abbb)$

$\rightarrow$ a aa abbb

$i \leq j$ Rejected as aaaabbb $\notin L$

The following language is irregular.

Always use min string of xyz

Prove that $L = \{ 0^{j^2} \mid j \geq 1 \}$ is not regular

$j = 1$ $\quad 0^{1^2} \rightarrow 0^1 \rightarrow 0$

$j = 2$ $\quad 0^{2^2} \rightarrow 0^4 \rightarrow 0000$

$j = 3$ $\quad 0^{3^2} \rightarrow 0^9 \rightarrow 000000000$

Choosing 0000 $\rightarrow$ $00(0)0$

$x\ y\ z$

$\rightarrow k = 2$

$00(00)0$

00000 (X)

Hence Language is not regular

$L_3 = \{ ww \mid w \in \{0,1\}^* \}$

$(0+1)^* \rightarrow$ ^ , 0, 1, 00, 01, 10, 11, 000, 010, 100, 110, 111, 001, 011, 101

$w = \wedge$ $\quad ww = \wedge$

$w = 0$ $\quad = 00$

$w = 1$ $\quad = 11$

$w = 00$ $\quad = 0000$

$w = 01$ $\quad = 0101$

$w = 11$ $\quad = 1111$

$w = 000$ $\quad = 000000$

$w = 010$ $\quad = 010010$

$w = 100$ $\quad = 100100$

$w = 110$ $\quad = 110110$

$w = 111$ $\quad = 111111$

w = 001 (001001) w = 10 101101

w = 01 011011

$00(10)^2$ → $00(10)^1 01$

00 101001

not part

Pick: Some Strings will not prove that a language is not regular for that use a string that proves it that a language is not regular

e.g $L = \{w \in (0,1)^* \mid w \text{ is a palindrome}\}$.

Check if L is regular or not.

$\wedge$, 0, 1, 00, 01, 10, 11, 000, 010, 100, 110, 111, 001, 011, 101, 0000, 0100, 1001, 1100, 1110, 0011, 1010, 1111, 0001, 0101, 1001, 1101, 0111, 1011

w = ^, 0, 1, 00, 11, 000, 111, 010, 101,

using 0110 Let say p=3 $|xy| <= 3$

x y z

0110

$0(1)^2 10$

0 11 10

Choosing another

0 1 0
↓ ↓ ↓
x y z

$0(1)^2 0 \rightarrow 0110 \in L$ ✓

Using

0 1 1 0
↓ ↓ ↓
x y z

$01(1)^2(0) \rightarrow 0110 \in L$